DHRUVA: 5-Day Upgrade Plan

Sep 30, 2026 · @nim

Priorities

The deadline moved to 5 October, which gives five working days. Spend them on the brief's unanswered questions first, because judges score against the brief, then on turning labelled test inputs into real data.

#	Upgrade	Owner	Why it scores	Size
1	Query #7: why fish productivity declined	Coder	Hardest query in the brief; no competitor answers it	1.5 days
2	Live IMD cyclone bulletins	Coder	Turns "test input" into "live" on the data table	1 day
3	Remaining brief queries	Coder	Covers all 8 example queries	1.5 days
4	Cited safety limits	Coder + team	Removes the most likely judge question	0.5 day
5	Polish and fixes	Coder	Nothing broken or half-translated on video	0.5 day
6	Field test, native-speaker check, video	Rest of team	Real users are something competitors can't copy in 5 days	In parallel

Rule for every upgrade: the same standards as the rest of DHRUVA. The rule engine decides; every number carries a source and time; anything simulated is labelled; missing data gives No data, never Safe. If an upgrade can't meet that by day 4, cut it and list it as future scope.

Where DHRUVA stands against the brief

DHRUVA answers 2 of the brief's 8 example queries fully and 3 partly; 3 are missing. Every row below should be "Done" by 4 October.

#	Example query in the brief	Today	Upgrade
1	Nearest Potential Fishing Zone today	Done	—
2	Is it safe to venture tomorrow morning?	Done	—
3	Tide, weather and sea conditions near my location	Partly: no tides	3
4	Lightning or cyclone alerts in my area	Partly: cyclone is test input, no lightning	2 and 3
5	Regions with high chlorophyll and favourable SST	Map layer only, no spoken answer	3
6	Safest route considering weather and sea state	Missing	3
7	Why fish productivity declined in a region	Missing	1
8	Zones to avoid due to hazards or geofencing	Partly: boundaries on the map, no spoken answer	3

Each new query must also appear as a chip in the Ask sheet, in Tamil, Malayalam, Odia and Hindi, and in the planner's keyword fallback.

Upgrade 1: "Why did fish productivity decline here?"

Build query #7 as a spoken answer that compares the last 30 days at the boat's area with the same weeks in earlier years, then says what is unusual and where to try instead. It must never claim a cause it cannot show.

Data

- Chlorophyll: Copernicus GlobColour L4 NRT, daily, 4 km (`OCEANCOLOUR_GLO_BGC_L4_NRT_009_102`). It covers April 2023 to now, so the same 30-day window in 2023, 2024 and 2025 gives a 3-year baseline.
- Sea surface temperature: the SST source already used by Ocean Analytics, over the same windows.
- Optional: wave height from the ERA5 archive, to tell "rough seas kept boats in" apart from "less plankton".

Method

1. Take a box of about 20 × 20 km around the boat or landing centre.
2. For each variable: current 30-day mean, baseline mean and spread (same window, earlier years), and the change.
3. Flag a variable as unusual only when the change is larger than the baseline's year-to-year spread.
4. Pick the nearest INCOIS fishing-zone line, or failing that the nearest grid area where chlorophyll is at or above its baseline, as "where to try".
5. The rule engine picks the sentence template; nothing is free-written.

Answer shape (template, all four parts, each number with source and dates):

- **Normal here:** "At this time of year, chlorophyll is usually about X."
- **What changed:** "This month it is Y, lower than in the last 3 years. The sea is Z °C warmer than usual."
- **Possible reason:** "Less plankton near the surface usually means fewer small fish." Always say "possible", never "because".
- **What to do:** "The nearest fishing zone with better conditions is N nm to the south-east."

Honesty rules

- If fewer than 2 baseline years have data for the box (cloud gaps), answer No data for this question.
- Say plainly that catch data is not used: "This compares the sea, not the catch."
- Wording to avoid: "fish have left", "overfishing", any cause stated as fact.

Done when

- [ ] The question works by voice in Tamil and appears as an Ask chip in all four languages.
- [ ] The answer shows an evidence card per variable with current value, baseline, change, dataset and dates.
- [ ] The agent trace shows Marine Data Discovery, Ocean Analytics, Geospatial Reasoning and Reporting ran.
- [ ] Tests cover: a real drop flagged, a normal year not flagged, missing baseline gives No data, and the number check passes.
- [ ] One worked example (for instance off Chennai) is captured for the deck and the video.

Upgrade 2: live IMD cyclone bulletins

Replace the hand-placed test cyclone with IMD's own published bulletins, so the Warn boats view and the BLOCKED override run on real warnings whenever one is active. IMD has no JSON API, but RSMC New Delhi publishes each National Bulletin as a PDF at a

stable path (example).

Method

1. Poll the RSMC New Delhi bulletin listing every 30 minutes and download any new National Bulletin PDF.
2. Extract the text and parse: system name and type (depression, cyclonic storm and so on), centre latitude and longitude, issue time, forecast track points, and the fishermen warning paragraph with its sea areas and dates.
3. Store the PDF, its hash, the parsed fields and the source URL, so every warning in an answer links back to the exact bulletin.
4. Feed the parsed centre and warning areas into the existing override: BLOCKED inside the warning area, labelled with the bulletin number and time.
5. Keep the manual "place a cyclone" control, but rename it "Test input" and show it only in the shore console.

Honesty rules

- If parsing fails or a field is missing, show the bulletin link and "Official warning: read IMD bulletin", and treat the area as at least Caution. Never drop a warning because it didn't parse.
- A bulletin older than its validity is not used.
- When no system is active, say "No active IMD cyclone bulletin as of <time>", not "No danger".

Done when

- [ ] At least 5 archived bulletins parse correctly in tests, including Michaung (Dec 2023) and one depression.
- [ ] The Michaung replay reads its cyclone position through the same parser.
- [ ] The data table status for IMD changes from "Test input" to "Live (bulletin parser)".
- [ ] A deliberately broken PDF gives Caution with a link, not Safe.

Upgrade 3: the remaining brief queries

Five smaller answers close the rest of the brief. Build them in this order; the safest route is the one to cut if time runs short.

Order	Query	Data	Answer	Done when
1	Zones to avoid (hazards, geofencing)	Existing boundaries in PostGIS, active IMD warning areas	"Stay out of: warning area 40 nm east; boundary line 12 nm south." Nearest 3, with distance and direction	Spoken answer; warning areas listed before boundaries; boundaries still marked indicative
2	High chlorophyll and favourable SST regions	Copernicus chlorophyll and SST already on the map	Nearest areas where both are in range, next to the INCOIS zone for comparison	Evidence card shows both values and the ranges used; ranges have a stated source or are marked provisional
3	Thunderstorm and lightning risk	Open-Meteo forecast weather codes 95–99 and CAPE	"Thunderstorms likely after 3 PM" with a Caution verdict	Labelled "forecast, not observed lightning"; IMD nowcast listed as future scope
4	Tides	Open-Meteo marine `sea_level_height_msl`	High and low water times for the harbour today	Labelled "model estimate, not for navigation", as Open-Meteo itself warns

Order	Query	Data	Answer	Done when
5	Safest route	Wave grid, warning areas, boundaries	Waypoints from harbour to the fishing zone that avoid cells over the wave limit and all warning areas	Route drawn on the map with "straight-line reference, not navigation"; if no safe route exists, say so

For every row: add the question to the planner's keyword fallback, add an Ask chip in four languages, and add at least 3 tests, one of them for missing data.

Upgrade 4: cited safety limits

The wave limit (2 m) and wind limit (20 kt) are marked "provisional" today. A judge's first question will be "where do these numbers come from?", so every limit needs a named source before 4 October, or it keeps its provisional label.

Where to look, in this order

1. IMD's fishermen warning wording in real bulletins, including the Michaung ones already in the system. They state the wind speeds at which fishermen are advised not to venture. Quote the exact sentence and bulletin number.
2. INCOIS Ocean State Forecast and High Wave Alert pages, for the wave heights at which INCOIS issues alerts.
3. The WMO sea state code (named bands of significant wave height, such as "moderate" and "rough"). Use it only to name the bands in plain words, not as a safety limit by itself.

Rules

- Store each limit with its value, unit, boat class, source title, URL and date in one table, and show that source on the evidence card.
- Where only a national IMD value exists, use it for every region and say so.
- Do not pick numbers from memory, blogs or other teams' repos. If no source is found, keep "provisional".

Done when

- [ ] The evidence card for wind and waves shows the cited source, not "Rule WAVE-S1".
- [ ] The data table status for safety limits reads "Cited" or still says "Provisional", truthfully.

Upgrade 5: polish and fixes

These are small, but each one is visible on video or to a judge clicking around.

- [ ] Translate the remaining English UI in Tamil, Malayalam, Odia and Hindi modes: "Ask before you go", "Ask about the sea", "Tap and speak your question", "Leaving", "Now", "Tomorrow 5 AM", "Listen again", "Nearest harbour", "Nearest potential fishing zone", "Head back to", the fishing-zone warning box.
- [ ] Home screen: show the last verdict (colour, icon, time) instead of only the intro text, once the fisherman has asked once.
- [ ] Show "worked out on this phone with no network" only on satellite answers.
- [ ] Automatic voice: Blocked and Unsafe speak by themselves and repeat until acknowledged; confirm on a real phone in airplane mode.
- [ ] Planner: when Gemini returns 429, show "keyword planner" in the trace (it already falls back; make sure the label reads cleanly).
- [ ] Point check: the Hindi sample still contains "लहर की ऊँचाई 1.25 मीटर" in the fisherman sentence. Raw numbers belong in the evidence view, not the spoken verdict.
- [ ] Run the full test suite and update the test count in the deck and portal text.
- [ ] Screenshot every screen at 100% zoom after the fixes, for the deck and video.

Non-code work (rest of the team)

These run in parallel with the code and need no programming. The field test is the single most persuasive addition to the deck.

1. Field test with real fishermen (1–2 October)

- Visit a fishing harbour or landing centre in Tamil Nadu (for example Kasimedu, Chennai) or a local fishermen's cooperative.
- Show 3 screens without explaining them: Safe, Blocked and No data. Ask "What is this telling you to do?" and note whether each person got it right.
- Ask one question by voice in Tamil and let them hear the answer.
- Record: how many people, how many understood each screen without reading, and 2–3 short quotes. Photos or video only with clear permission, and no faces without consent.
- Put the result in one line on the deck, for example "Tested with N fishermen at <place>: M of N understood the Blocked screen without reading." Use the real numbers, whatever they are.

2. Native-speaker check (by 2 October)

- Ask a Tamil speaker (ideally from a fishing family) to read every Tamil verdict, reason and chip. Note wording that sounds unnatural and send the fixes to the coder.
- Repeat for Malayalam, Odia and Hindi if you can find speakers; otherwise keep "In review" for those.

3. Registrations and letters (by 3 October)

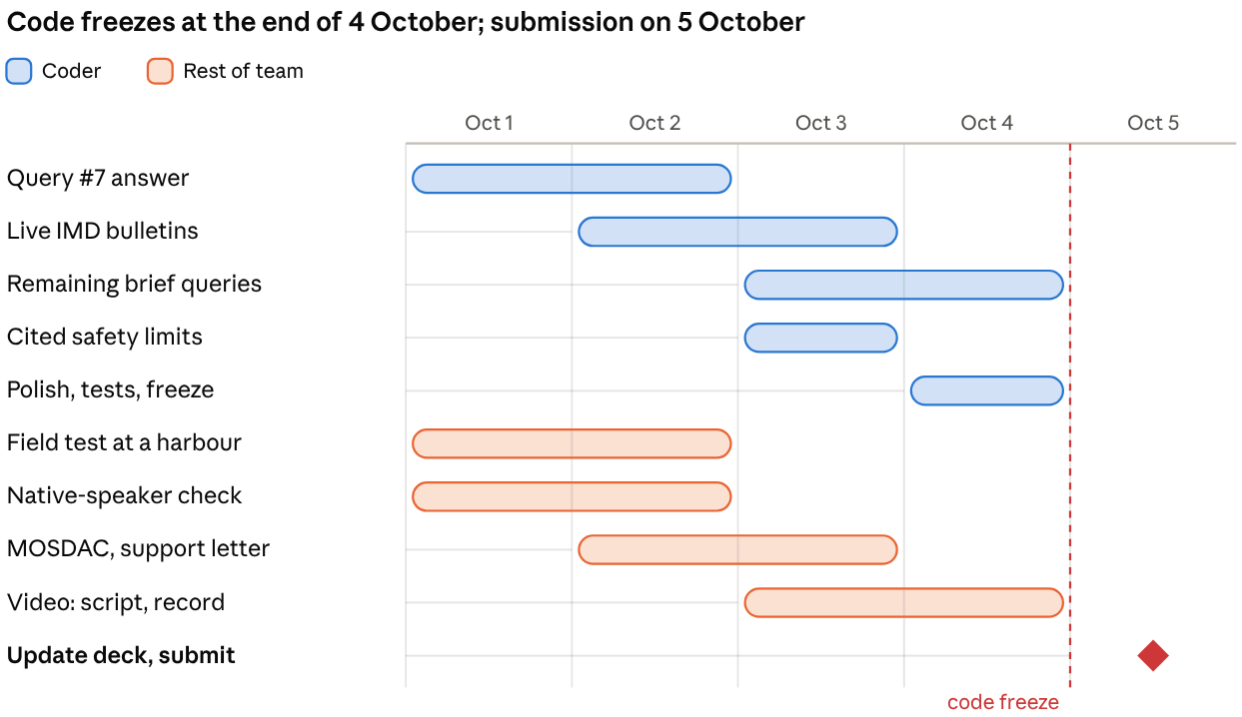
- Register on MOSDAC and request access to its ocean products. Screenshot the request.
- If the team's mentor or a fisheries officer will write a two-line note of support, attach it to the submission if the portal allows, or quote it on the impact slide.

4. Demo video (script 3 October, record 4 October)

- Under 3 minutes. Hook: "After Ockhi, India learned warnings must reach boats at sea. ISRO built that channel. Here's what DHRUVA would have sent each boat during Cyclone Michaung."
- Show in order: a Tamil voice question and spoken Safe answer; query #7; the Michaung replay in Warn boats; airplane mode with the Blocked alert speaking; the field-test result.
- Before recording: no broken map tiles, no English left in Tamil mode, every number plausible, simulated links labelled on screen.

Schedule

The coder builds the three upgrades back to back while the rest of the team runs the field test, checks, video and submission in parallel. No new features after 4 October: day 5 is only for updating the deck and portal text with real results and submitting.



5-day plan · coder and rest of team, 1–5 October

If day 2 slips, cut the safest route (last row of Upgrade 3) first, then thunderstorms and tides. Query #7 and the IMD bulletins are not cut.

Freeze checklist (evening of 4 October)

Nothing is submitted until every box is ticked or the item is moved to future scope on the deck.

Product

- [ ] All 8 brief queries answer by voice in Tamil, or the missing ones are listed as future scope.
- [ ] Every number on every screen has a source and time; no impossible values (for example SST outside 20–32 °C).
- [ ] Missing data shows No data; unknown alerts show at least Caution.
- [ ] Airplane mode on a real phone: the Blocked alert arrives, speaks by itself and repeats until acknowledged.
- [ ] No English left in Tamil mode; no broken map tiles or API-key watermarks.
- [ ] Simulated links, test inputs and provisional limits are labelled on screen.
- [ ] Full test suite passes; note the new test count.

Submission

- [ ] Deck updated: Team ID, new test count, query #7 example, field-test line, data table statuses (IMD, limits, wording).
- [ ] Portal text updated to match the deck, with no claim the deck doesn't make.
- [ ] Deck exported to PDF from PowerPoint, under 10 MB, 6 slides.
- [ ] Video under 3 minutes, uploaded and the link tested in a private window.
- [ ] One person who hasn't seen the work reads the deck and says what DHRUVA does in one sentence.

Sources

Pages opened on 30 September 2026 to check the data named in the specs.

- Open-Meteo Marine Weather API: `sea_level_height_msl` (tides), currents and SST; its note that tide accuracy is limited near coasts and not for navigation.
- Copernicus GlobColour L4 NRT, OCEANCOLOUR_GLO_BGC_L4_NRT_009_102: daily and monthly chlorophyll at 4 km, April 2023 to now.
- IMD RSMC New Delhi, National Bulletin No. 33, 29 Nov 2025: an example of the bulletin PDFs the parser reads.